



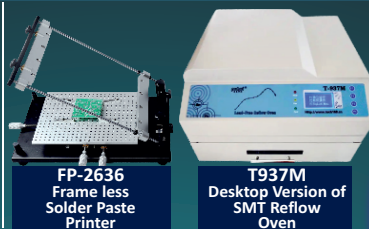
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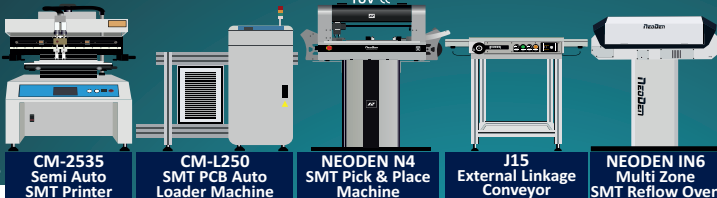
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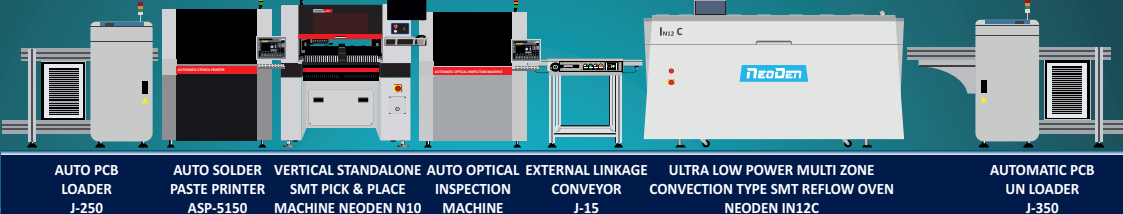
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industry has witnessed a transformative shift. These technologies offer powerful tools for optimising material properties, improving production efficiency, and ensuring product quality.

Materials design and synthesis

AI has revolutionised the materials design and synthesis phase by accelerating the introduction and predicting the interaction of novel materials with desired properties. Researchers can leverage AI algorithms to predict material properties based on chemical composition, crystal structure, structural interactions, and other physical and chemical factors. ML models trained on vast material databases can generate valuable insights, leading to the creation of new formulations tailored for specific applications.

Materials property prediction.

AI-driven materials property prediction models can significantly reduce the time and cost required for material

development. These models utilise data from various sources, such as quantum calculations, experimental data, and simulations, to predict material properties like mechanical strength, electrical and thermal conductivity, and thermal stability, etc. Researchers can use these predictions to identify promising candidates for further study, expediting the materials formulation and synthesis process. This can significantly reduce the time to arrive at the right material uses and avoid the trial and error method in the laboratory.

Materials informatics. Materials informatics combines AI and ML techniques to extract valuable information from vast datasets. By analysing data related to materials' synthesis methods, properties, and performance, researchers can uncover hidden correlations and patterns. This knowledge helps in optimising material manufacturing processes,

ensuring better performance and resource utilisation.

Process optimisation and quality control

In material manufacturing, the production process and product quality are of paramount importance. AI and ML technologies enable real-time monitoring and control of manufacturing processes, ensuring consistency and quality.

Predictive maintenance. Machine learning models can predict equipment failures and maintenance requirements, allowing manufacturers to schedule maintenance activities proactively. This minimises unplanned downtime, reduces production losses, and extends the lifespan of machinery.

Quality assurance. AI-powered image recognition and computer vision systems are employed to inspect and assess the quality of materials dur-